Quantum-Safe Gateway

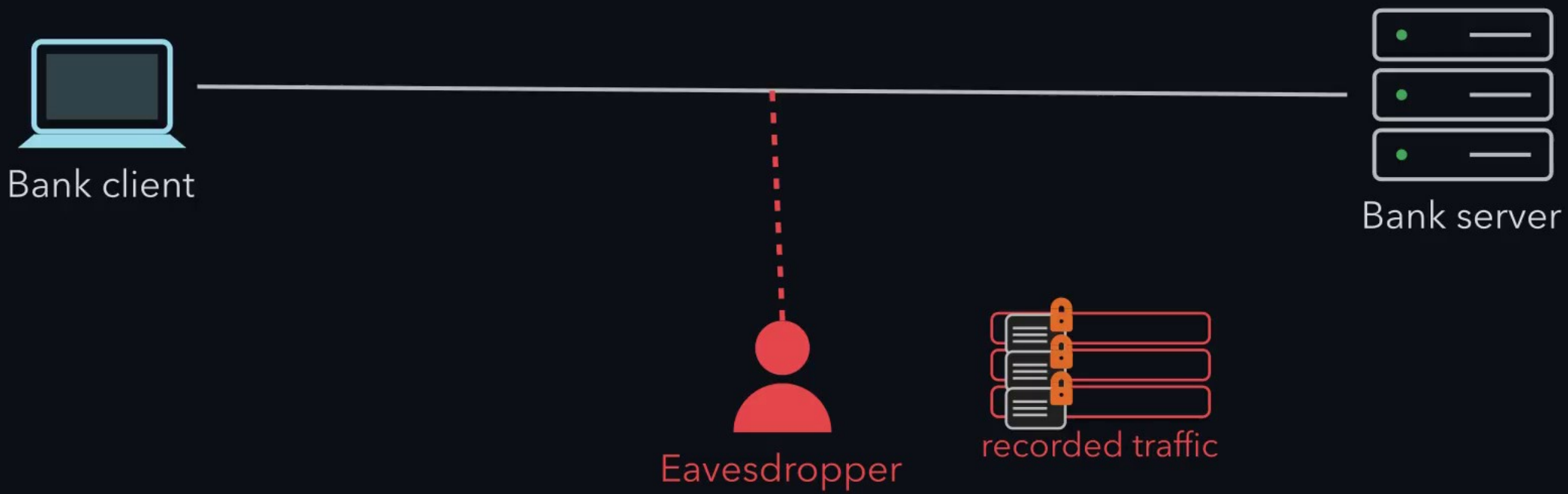
for legacy systems

IBM Z Datathon 2026 · Theme 2: AI Secured



Harvest now, decrypt later

year
2026



Today: recorded, but still locked

Harvest now, decrypt later

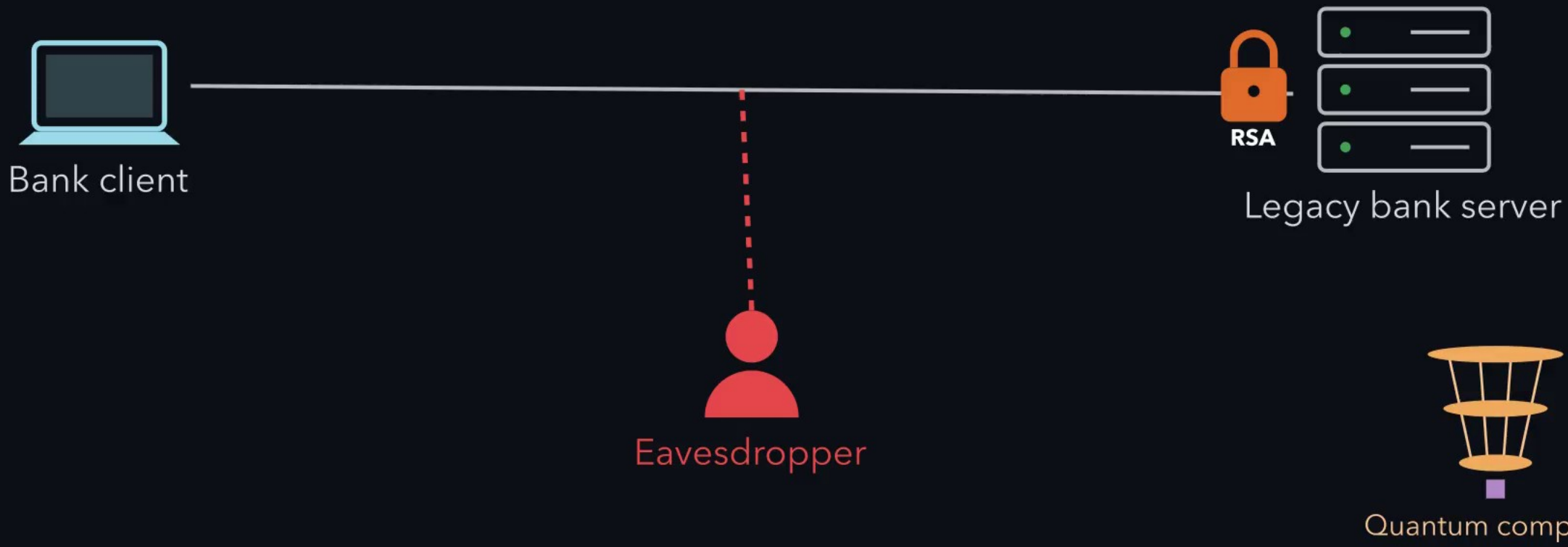
year
2036



Data that must stay secret for 10+ years is already at risk

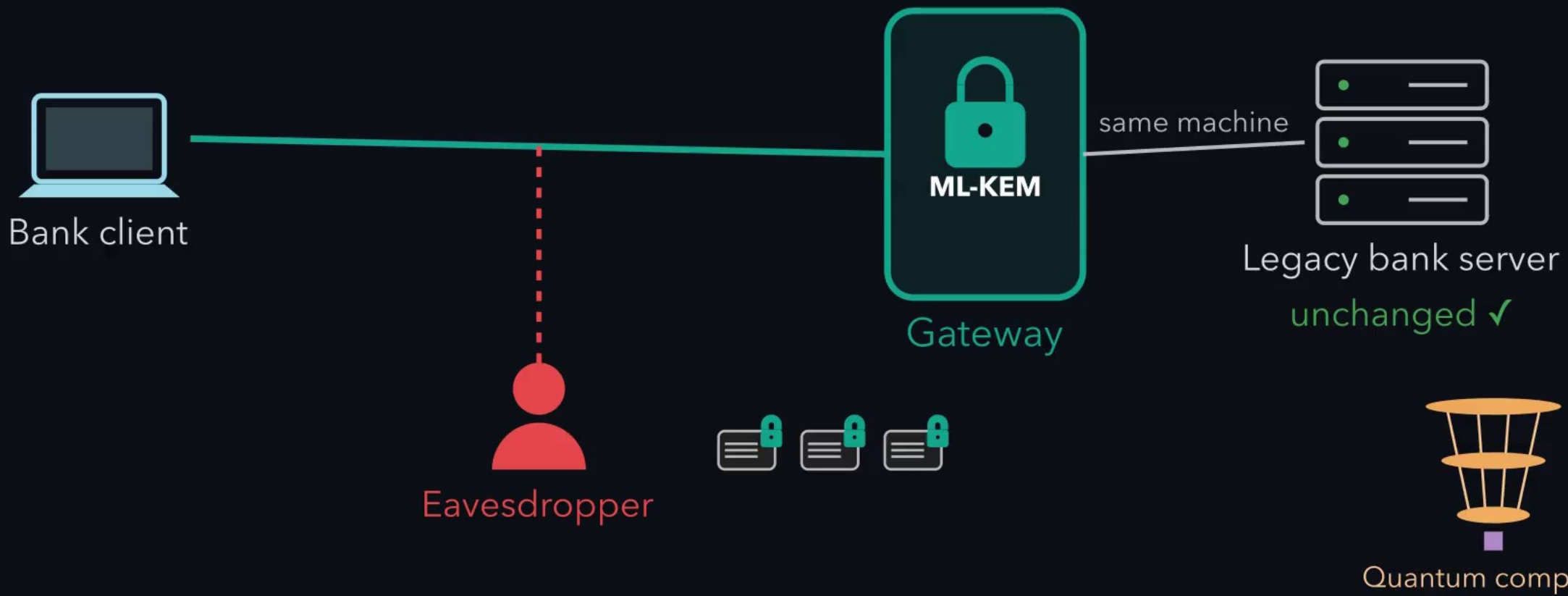
Illustration: quantum computers able to do this do not exist yet.

A drop-in quantum-safe gateway



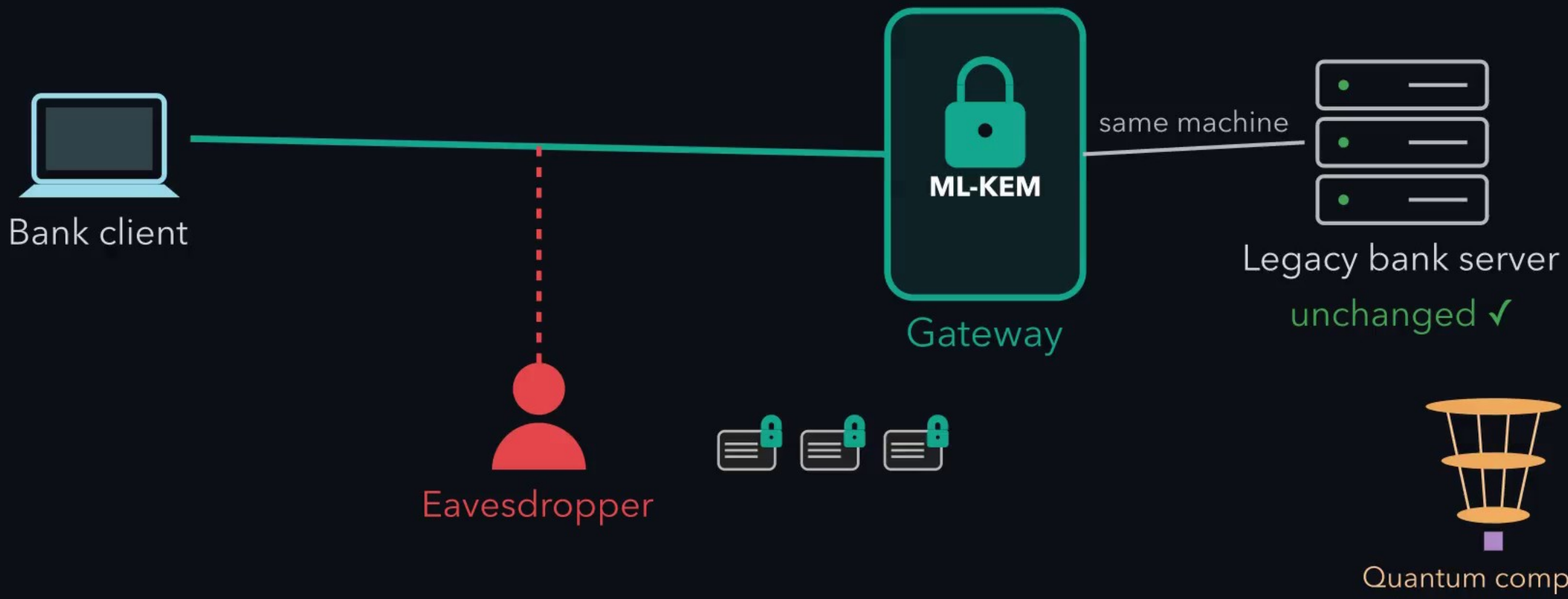
Rewriting old systems is slow, expensive and risky

A drop-in quantum-safe gateway



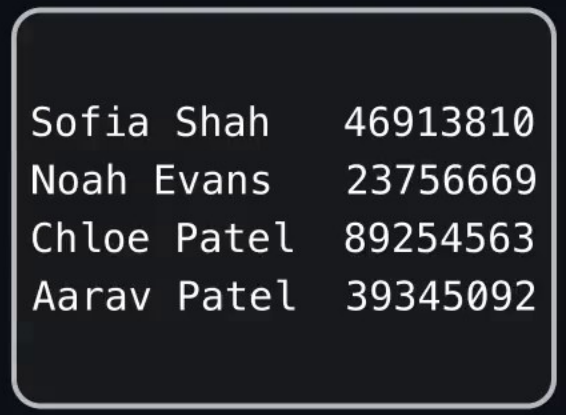
…and the recordings stay locked

A drop-in quantum-safe gateway



Strict mode: no quantum-safe lock, no connection

Data at rest



bank_records.json

A thief who copies the disk gets everything

Data at rest



Step 1: a fresh AES-256 key scrambles the file

Data at rest



Only the private key can unlock it; change one byte and unlock refuses

Downgrade attack



Client



Attacker in the middle



Gateway

"I can use:"

- `X25519MLKEM768` quantum-safe
- `X25519` classical

The client offers the quantum-safe lock, and the old one as a fallback

Downgrade attack



Client



Attacker in the middle



Gateway

"I can use:"
- X25519 classical

The attacker deletes the quantum-safe option

Downgrade attack

ATTACK DETECTED: PQC downgrade



Client



Attacker in the middle



Gateway

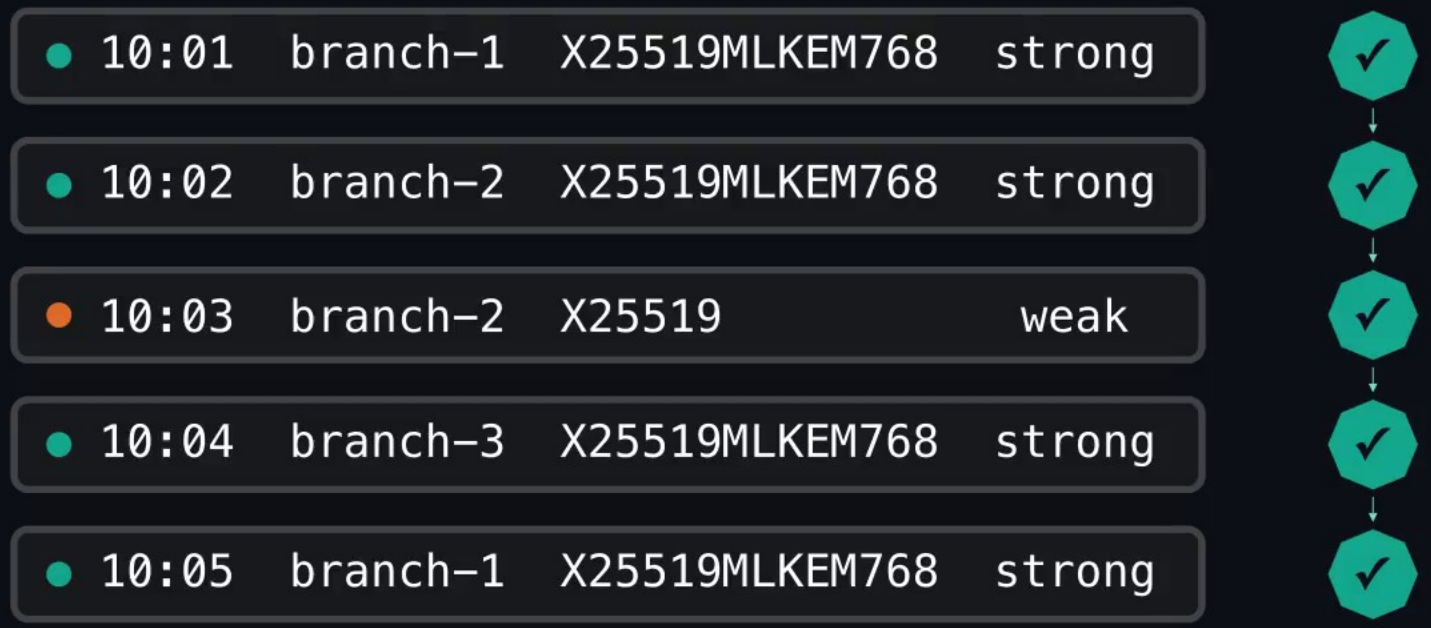
AI tripwire

this client's handshakes



The gateway refuses it, and the AI flags it

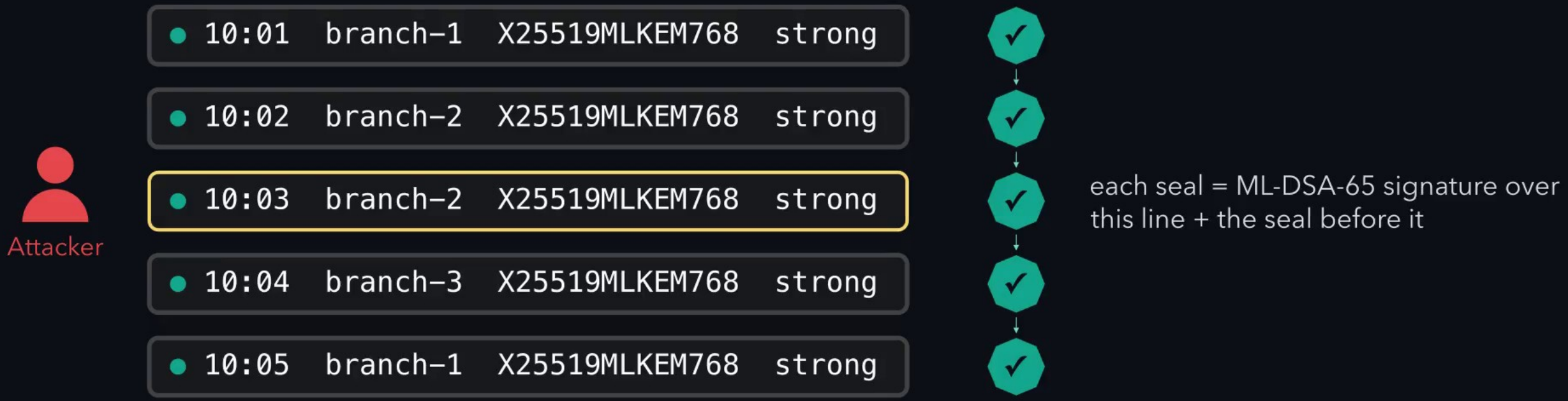
Tamper-proof audit log



each seal = ML-DSA-65 signature over this line + the seal before it

Every connection is written down and sealed

Tamper-proof audit log



The attacker edits line 3 to hide the downgrade

Tamper-proof audit log



Edits, deletions, insertions and reordering are all caught

Running on IBM LinuxONE (s390x)



+0.33 ms per connection to be quantum-safe

Throughput: 2,371 → 2,224 MB/s (−6%), same AES bulk cipher

Measured on the team's LinuxONE VM (3 CPUs, loopback) · gateway/bench/benchmarks.md

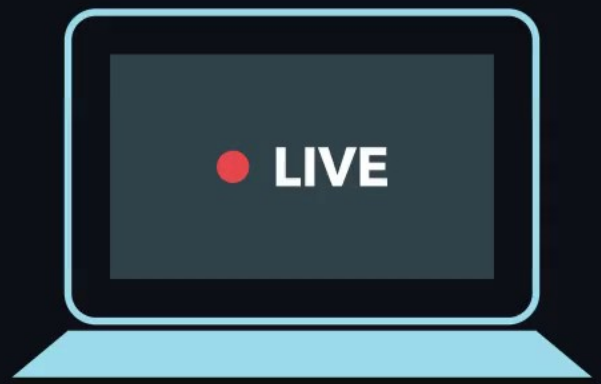
Quantum-safe, without rewriting

- ✓ Hybrid post-quantum TLS — X25519 + ML-KEM-768
- ✓ Data at rest — AES-256-GCM, keys wrapped with ML-KEM
- ✓ AI downgrade tripwire — rule + kNN anomaly model
- ✓ Tamper-proof logs — ML-DSA-65 signature chain
- ✓ Legacy server — unchanged

Built and benchmarked on IBM LinuxONE · IBM Z Datathon 2026

Here it is running

The real dashboard and detector, with simulated bank traffic



Quantum-Safe Gateway: Downgrade & Anomaly Tripwire

updated 19:00:57

Reading `connections.jsonl`. Layer 1: deterministic PQC-downgrade rule. Layer 2: kNN novelty model trained on normal traffic.

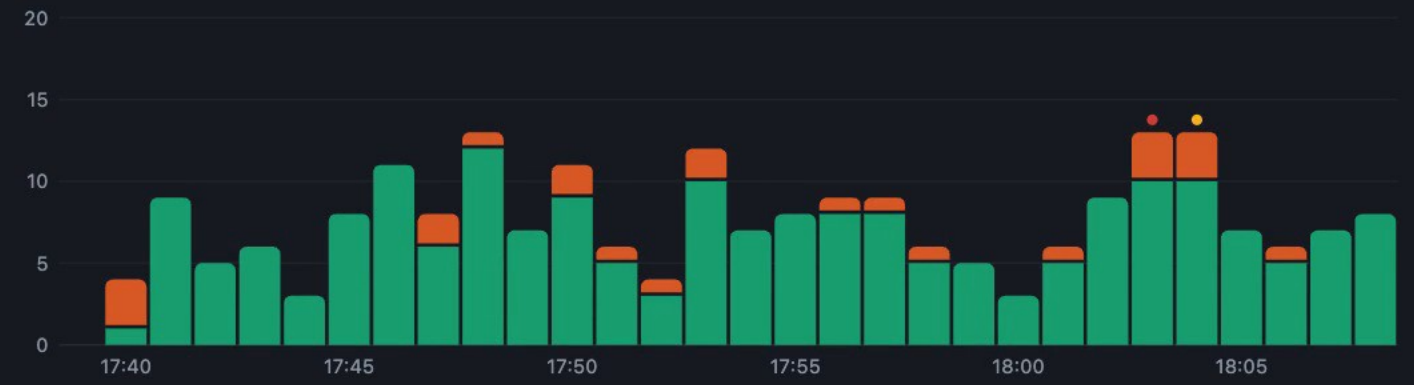
SECURE: all recent handshakes look normal

Connections checked	Quantum-safe share	Harvest-exposed connections	Alerts	Clients needing action	Log integrity (ML-DSA seals)
223	90%	23	3	11 / 45	✕ TAMPERED
8.2 per minute (last 5 min)	of 223 completed handshakes	classical-only key exchange: could be recorded now and decrypted later	1 downgrade · 2 AI anomaly	1 downgraded · 1 anomalous · 9 need PQC upgrade	line 173: content was changed after it was sealed

Traffic per minute

Quantum-safe · Classical · Refused by gateway · Downgrade · Anomaly

Markers above a column: alerts in that minute



Key exchange negotiated

Post-quantum · Classical

Completed handshakes



AI anomaly score, last 150 connections

Normal · AI anomaly · Downgrade rule

Distance from normal traffic (log scale). Above the line, the model raises an alert.



Alerts by type

Rule · AI model

Main reason per alert

